

Cleaning out the garage one warm day, I stumbled upon an ancient artifact, a big white box covered in dust—a computer from 2000. I spent hours taking it apart and trying to understand its inner workings, examining each component with awe. The blinking lights, the intricate circuitry, and the wonder of it captured my imagination.

My fascination with computers didn't end in the garage. In 9th grade, I joined my high school's robotics team, strengthening my desire to pursue computer engineering. While on the robotics team, I gained valuable knowledge of motors, wiring, and programming and the importance of attention to detail where even a tiny electrical or programming error can cause an entire subsystem or even the entire robot to fail. The enjoyment I found in programming and working on the robot's electrical systems is what led me to pursue a computer engineering degree at the University of Maryland (UMD).

In the summer of my junior year and throughout my senior year of high school, I participated in two internships at the Johns Hopkins Applied Physics Laboratory. During my first internship, I worked on the navigation software for Dragonfly, an APL-led initiative to send an octocopter to the moon Titan in search of past or present life in 2027. With no maps of Titan, Dragonfly must navigate using landmarks and be able to compare current and past images to determine its location. My primary task was to compare the effectiveness of two camera calibration systems—the MATLAB Camera Calibrator and the OpenCV camera calibration software for improved landmark correlation. After calibrating the Dragonfly prototype camera using a checkerboard I used the generated camera parameters and used MATLAB for the flight simulations. As a result of my research, Dragonfly's navigation team now uses the MATLAB Camera Calibrator for prototype test flights.

During my second internship, my main task was to train the YOLOv8 computer vision model to accurately detect lunar craters. However, many existing lunar crater datasets lacked diversity, with insufficient non-angled photos. I used MATLAB to generate a dataset of lunar craters using a digital elevation map of a section of the moon. I annotated these images and used them as training data to train the model in Python, which led to a significant improvement in the computer vision model's detection accuracy.

Both of my internships gave me a greater appreciation of the application of computer engineering to the real world.

Becoming part of the ANCHOR Fellowship program would be an exciting opportunity to learn about new technologies. I am particularly drawn to the research topics of unmanned underwater vehicles (UUVs) and systems engineering/modeling.

Having previously worked on surface-based robotics, I'm eager to explore robotics in a different domain through UUVs. I'm excited to learn about the unique challenges presented by a marine environment. Specifically, I'm interested in autonomous perception and distributed intelligence. The ability to develop robots that can navigate by themselves with little to no human assistance is something that fascinates and motivates me to pursue UUV research.

I am also interested in delving into the realm of cybersecurity because of the constantly evolving nature of the field. I'm excited to apply my mind to new and unexplored challenges and learn more about cybersecurity overall. In cybersecurity, I'm drawn to embedded systems security because of the critical role it plays in the control of autonomous vehicles. I've always been interested in cracking codes so a field like cryptography which is expressly about cracking codes is also interesting to me.

The ANCHOR Fellowship represents an exciting opportunity to develop my skills, contribute to impactful research, and further my understanding of the various fields within computer engineering. I look forward to delving into undersea vehicle research and gaining industry-relevant skills and experience. I am also excited to meet other students and faculty in the ANCHOR Fellowship Program and develop professional and personal connections.

In the future, I aim to work in a computer engineering-related job in the defense industry, applying the knowledge and experience I've gained to save lives and strengthen our national security. I also plan to earn a master's degree to deepen my understanding of computer engineering and further investigate the topics that I am interested in. After a few years working in the industry, I would like to pursue a Master's in Business Administration. This will give me the knowledge and expertise needed to lead teams in developing technologies that make the world a safer place.

By participating in the ANCHOR Fellowship, I aspire to learn more about the field of computer engineering and gain the skills I need to turn innovative ideas into practical solutions.